Assessing Anthropogenic Pollution in Sodus Point, NY

Marion Eighth-Grade & Twelfth-Grade Students
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Introduction

Sodus Point, located on the southeastern shore of Lake Ontario, is a lakeside community known for its rich biodiversity and recreational boating. Surrounded by the newly designated Lake Ontario Marine Sanctuary and Sodus Bay, it faces increasing environmental pressures including pollution that pose significant threats to water, sand, and aquatic quality on the macro and micro scale.

Hypothesis

There is a significant presence of anthropogenic (man-made) materials within Sodus Bay and Lake Ontario's waters, shores, and fish, presenting potentially harmful effects to the health of these environments and the communities that surround them.

Methods

Sand samples were collected from both the bayside and lakeside, with further geographical distinction from nearest to the shoreline and nearest to the parking lot. The sand was then sifted through a series of sieves, with mesh sizes 2 mm, 177 µm, and 74 µm. Particles retained from the 74 µm and 177 µm sieve were transferred to a one-liter jar and mixed with sodium chloride (NaCl) solution to perform density separation. After 24 hours, the separation was filtered with a funnel and 30 micron pore Whatman paper filter, and the retained debris was then collected by pressing clear sticky tape along the surface of the filter and onto a microscope slide.

Water samples were gathered at multiple points along the length of the pier in both Sodus Bay and Lake Ontario. Two buckets were used during sampling: one for rinsing hands and jars, and the other to collect the sample water. The water samples were filtered and processed using the same procedure as the sand samples, described above.

All the fish, of the species *Perca flavescens* (yellow perch), were collected from the bayside. The fish guts were placed in potassium hydroxide (KOH) to dissolve organic material, leaving behind inorganic material. The residue was then analyzed using density separation followed by microscopy according to the preceding established procedures.

Litter data was collected along the shores of Sodus Bay and Lake Ontario using the *Rubbish - Love Where You Live* App. To log data, participants took a picture of litter then categorized it. Data analysis was performed on this dataset.

Results

Beach sand was sampled at 16 different locations and a total of 43 anthropogenic materials were found. Of these, 22 came from the lake shore (8 fibers, 1 sphere, and 13 fragments). The remaining 21 anthropogenic items were found on the bay shore (9 fibers and 12 fragments).

Water was sampled at 8 different locations and a total of 28 anthropogenic materials were found. In Lake Ontario, 11 particles were found (5 fibers and 6 fragments). The remaining 17 were found in the bay (12 fibers and 5 fragments). The bayside samples showed a higher fiber count compared to the lakeside water samples.

Anthropogenic materials were found in every yellow perch dissected (8 fish) and a total of 34 anthropogenic materials were found (22 fibers and 12 fragments).

During the lake shore cleanup, 275 total pieces of litter were tracked and categorized with the Rubbish app (242 pieces of plastic, 15 pieces of miscellaneous litter, 11 pieces of paper, 6 cigarette butts, and 1 fiber). During the bay shore cleanup, 131 total pieces of litter were tracked and categorized (77 pieces of plastic, 28 pieces of tobacco, 8 pieces of paper, 6 pieces of metal and miscellaneous litter each, 4 fibers, and 1 food scrap and glass each).

Sand Sample Analysis for the Presence of Anthropogenic Materials

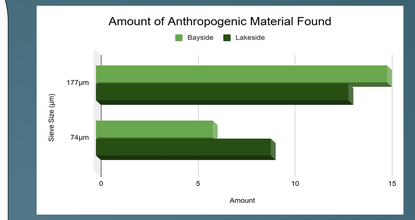


Figure 3: Total amount of anthropogenic materials found within lakeside and bayside sand samples per sieve size.



Image 3: Stack of sieves used.

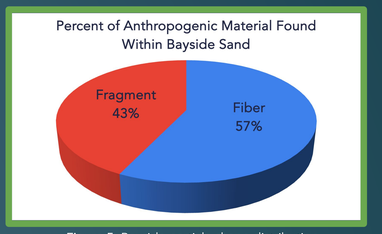


Figure 5: Bayside particle shape distribution.



Image 4: Eighth grade students and sustainability fellow, Sadie, filtering sand.

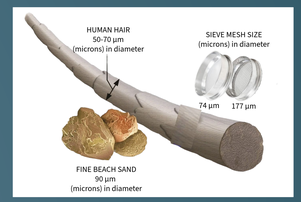


Figure 4: Size comparison of the diameter of a human hair and a particle of fine beach sand.

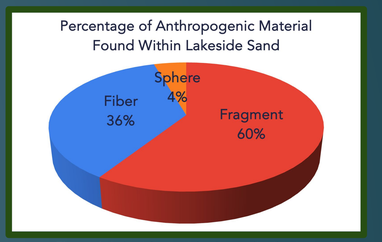


Figure 6: Lakeside particle shape distribution.

Litter Data Characterized via the Rubbish Application

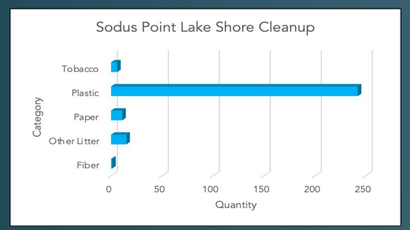


Figure 1: Type and quantity of litter tracked on the shore of Lake Ontario.

Category	Quantity
Fiber	1
Other Litter	15
Paper	11
Plastic	242
Tobacco	6
Total	275

Table 1: Lake shore litter cleanup composition totals.

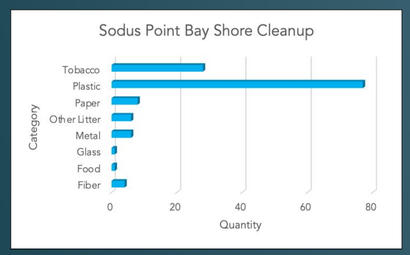
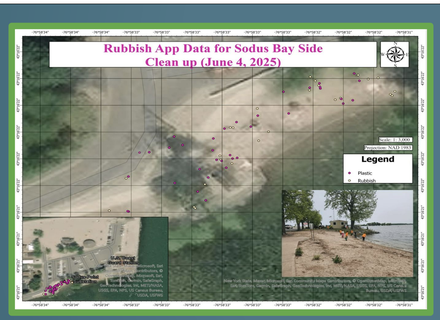


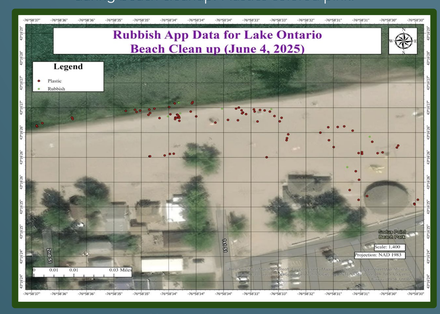
Figure 2: Type and quantity of litter tracked on the shore of Sodus Bay.

Category	Quantity
Fiber	4
Food	1
Glass	1
Metal	6
Other Litter	6
Paper	8
Plastic	77
Tobacco	28
Total	131

Table 2: Bay shore litter cleanup composition totals.



Map 1: Location of litter collected on the Sodus Bay shore during beach cleanup. Plastics colored pink.



Map 2: Location of litter collected on the Lake Ontario shore during beach cleanup. Plastics colored red.

Download the Rubbish app on your phone!

If you have the Rubbish app, scan this QR code to join our Community Cleanup team!



Image 1: 12th graders and their teacher Dr. P, using Rubbish to collect litter data in Sodus Point.



Image 2: 8th graders collecting litter data on the shore of the Sodus Bay.

Perca flavescens Intestine Analysis for the Consumption of Anthropogenic Materials

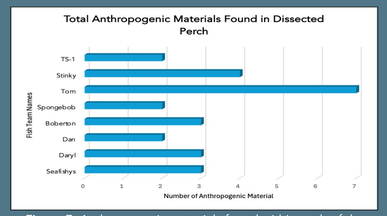


Figure 7: Anthropogenic materials found within each of the 8th grade groups' *Perca flavescens* digestive tract sample.

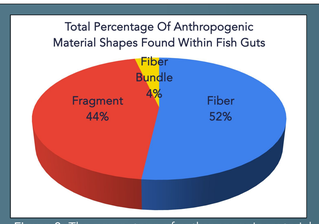


Figure 8: The percentage of anthropogenic material shape found within 8th graders' *Perca flavescens* digestive tract samples.

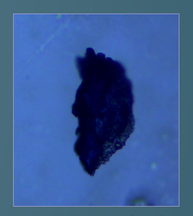


Image 5: Fragment found in dissected *Perca flavescens*.

Average # of Anthropogenic Material Found per Fish	Average # of Anthropogenic Material Found per Gram of Stomach	Average # of Anthropogenic Material Found per Gram of Total	Average Stomach Weight to Total Weight
3.78	0.21	0.02	11.85%

Table 3: Averages of plastic and anthropogenic material within dissected *Perca flavescens* and average stomach to total weight ratio.



Image 6: Eighth grade students and sustainability fellow, Eileen, dissecting *Perca flavescens*.



Image 7: Fiber found in dissected *Perca flavescens*.

Water Sample Analysis for the Presence of Anthropogenic Materials



Image 8: Students filtering water samples

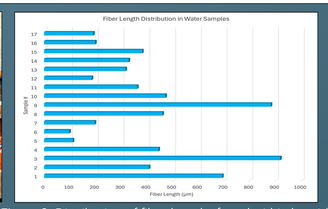


Figure 9: Distribution of fiber lengths found within bayside and lakeside.

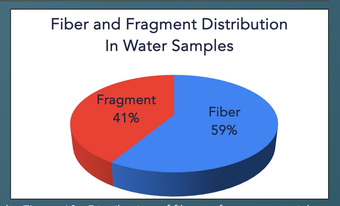


Figure 10: Distribution of fiber vs fragment particle shapes found within bay and lake water.



Image 9: Fibers found in bayside water sample.

Average Fiber Length:	Maximum Fiber Length:	Minimum Fiber Length:
407.14 µm	916.85 µm	98.03 µm

Table 4: Average, maximum, and minimum of bayside and lakeside fiber lengths.

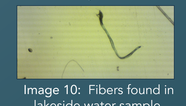


Image 10: Fibers found in lakeside water sample.

Discussion

Our results indicate a significant presence of anthropogenic materials throughout Lake Ontario and Sodus Bay shores, water, and aquatic organisms. A higher percentage of fibers than fragments in water samples support the inference that bathing suits shedding and laundry waste water are two key sources of the pollution in Sodus Bay and Lake Ontario. Because microplastics were found in the digestive tracts of the associated perch living in Sodus Bay, it can be inferred that microplastics have the capability to travel through water, air, and land movement and travel within organisms, particularly through normal food chain behavior of consumption and digestion. Further research should be conducted to determine if this capability extends to humans who consume fish and the consequential health impacts. This poses the critical question: is our community safe from microplastics? As many community members actively fish in the bay and either consume or sell their catch, the potential risk of consuming microplastics and anthropogenic material impacts hundreds of community members within Sodus Bay.

Future Directions

- Investigate the presence of microplastics within the fillet of yellow perch, a common cut of fish meat consumed by the community.
- Identify the most common types of plastic found using Raman Spectroscopy to identify the plastic type and potential sources.
- Investigate the presence of microplastics within community members who consume fish from Sodus Bay.
- Further explore why there might be more diverse anthropogenic material colors in the bayside compared to the lakeside.
- Work with community members, businesses, and governments to come up with solutions for mitigating anthropogenic pollution in Sodus Bay.

References

1. Chen, Q., Shi, G., Revell, L. E., Zhang, J., Zuo, C., Wang, D., Le Ru, E. C., Wu, G., & Mitrano, D. M. (2023, November 30). Long-range atmospheric transport of microplastics across the Southern Hemisphere. Nature News. https://www.nature.com/articles/s41467-023-43695-0